

# Saarang Panchavati, Research Scientist

 saarangp |  saarangp |  saarangp.github.io |  saarang@g.ucla.edu

## SUMMARY

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I recently completed my PhD in Medical Informatics at UCLA, where I built self-supervised foundation models for EEG and studied how seizure networks evolve through thalamocortical circuits. My current work combines both threads into a world model that predicts brain-state trajectories under stimulation for closed-loop neurostimulation planning. Broadly, I am interested in representation learning for biological signals and in mechanistic interpretability – understanding what these models learn about physiology and how that can drive clinical insight. I previously worked with Meta FAIR’s Brain & AI group on multimodal neural encoding and decoding. Published at ICLR, Epilepsia, and ICASSP.

## EDUCATION

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### University of California, Los Angeles

Sep 2021 – May 2026

Ph.D. in Medical Informatics

*Dissertation: Learning Pathological Brain Dynamics: EEG Foundation Models and Epilepsy*

*Advisor: William Speier*

*NSF GRFP Fellow*

*Training in Neurotechnology Translation Fellow*

### University of California, Berkeley

May 2021

B.S. in Bioengineering & Electrical Engineering & Computer Science

*Sutardja Dai Entrepreneurship Fellowship*

## SELECTED PUBLICATIONS

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Banville, H., S. d’Ascoli, S. Dahan, J. Rapin, M. Careil, Y. Benchetrit, J. Lévy, **Panchavati, Saarang**, A. Ratouchniak, E. Cascardi, et al. (2026). “NeuralBench: A Unifying Framework to Benchmark NeuroAI Models”. *arXiv preprint arXiv:2605.08495*.

Ivezić, V., M. Pleasure, A. Radhachandran, **Panchavati, Saarang**, S. Athreya, V. Sant, et al. (2026). “Pretext Matters: An Empirical Study of SSL Methods in Medical Imaging”. *arXiv preprint arXiv:2603.22649*.

**Panchavati, Saarang**, U. Panchavati, C. Arnold, and W. Speier (2026). “Laya: A LeJEPa Approach to EEG via Latent Prediction over Reconstruction”. *arXiv preprint arXiv:2603.16281*.

**Panchavati, Saarang\***, A. Daida\*, S. Kanai, S. Oana, H. Ono, M. Izumi, et al. (2026). “Distinct Spectral and Directional Thalamocortical Network Dynamics Define Focal Seizure Evolution”. *medRxiv*.

Daida, A., **Panchavati, Saarang**, S. Oana, S. Kanai, Y. Zhang, Y. Ding, et al. (2025). “Evidence of thalamocortical network activation during epileptic spasms: A thalamic stereotactic EEG study”. *Epilepsia*.

**Panchavati, Saarang**, C. Arnold, and W. Speier (2024). “Mentality: A Mamba-based Approach towards Foundation Models for EEG”. *The Twelfth International Conference on Learning Representations (ICLR)*.

**Panchavati, Saarang**, A. Daida, B. Edmonds, M. Miyakoshi, S. Oana, S. S. Ahn, et al. (2024). “Uncovering spatiotemporal dynamics of the corticothalamic network at ictal onset”. *Epilepsia* 65.7, pp. 1989–2003.

Parthasarathy, N., J. Soetedjo, **Panchavati, Saarang**, N. Parthasarathy, C. Arnold, N. Pouratian, et al. (2024). “High performance P300 spellers using GPT2 word prediction with cross-subject training”. *Brain-Computer Interfaces* 11.4, pp. 210–224.

**Panchavati, Saarang**, S. Vander Dussen, H. Semwal, A. Ali, J. Chen, H. Li, et al. (2023). “Pretrained transformers for seizure detection”. *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, pp. 1–2.

**Panchavati, Saarang**, N. S. Zelin, A. Garikipati, E. Pellegrini, Z. Iqbal, G. Barnes, et al. (2022). “A comparative analysis of machine learning approaches to predict C. difficile infection in hospitalized patients”. *American Journal of Infection Control* 50.3, pp. 250–257.

Pellegrini, E., **Panchavati, Saarang**, C. Lam, A. Garikipati, N. Zelin, G. Barnes, et al. (2021). “A Machine Learning Clinical Decision Support Tool For Myocardial Infarction Diagnosis”. *Journal of the American College of Cardiology* 77.18.Supplement\_1, pp. 653–653.

Mohamadlou, H., **Panchavati, Saarang**, J. Calvert, A. Lynn-Palevsky, S. Le, A. Allen, et al. (2020). “Multicenter validation of a machine-learning algorithm for 48-h all-cause mortality prediction”. *Health Informatics J.* 26.3, pp. 1912–1925.

Find more on [Google Scholar](#)

## EXPERIENCE

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### Medical Informatics PhD Researcher

September 2021 – May 2026

*Biomedical AI Research, UCLA, Los Angeles*

- Developed Laya, a JEPA-based EEG foundation model pretrained on 30K hours across 20K+ subjects, achieving state-of-the-art clinical EEG representations under frozen linear probing.
- Designed cross-dataset evaluation suites assessing transfer, noise robustness, and low-label adaptation across heterogeneous clinical cohorts.
- Led translational research in pediatric epilepsy: characterized thalamocortical seizure dynamics using SEEG in 19 patients, identifying frequency-specific biomarkers that classify seizure state (AUROC 0.83–0.84).
- Built a seizure world model that predicts brain-state trajectories under candidate stimulation, enabling counterfactual rollouts for closed-loop neurostimulation planning.
- Mentored undergraduate researchers to build a real-time motor-imagery BCI pipeline (PyTorch, MNE, BCI2000), culminating in a first-place award at UCLA Bioengineering Research Day.

### Research Scientist Intern

Mar 2025 - August 2025

*Meta FAIR (BrainAI), London*

- Conducted research on multi-modal representation learning for neural decoding and encoding in the Brain AI group across diverse data types (neural data, imaging, vision, language).
- Contributed to scalable open-source neural data training and evaluation pipelines in large collaborative codebases (neuralnet, neuralbench); developed benchmarks to assess cross-task and cross-modal transfer

### Data Science Intern

March 2020 – May 2022

*Dascena, Remote*

- Evaluated deep learning models for ICU C. difficile prediction, contributing to a peer-reviewed publication.
- Built longitudinal EHR predictive models across multiple clinical indications, resulting in three peer-reviewed publications.
- Developed EHR data-processing pipelines using AWS and SQL to support model development and retrospective analysis.

### Undergraduate Researcher

January 2018 – May 2020

*Maharbiz Lab, Berkeley*

- Designed experiments to evaluate neural-implant biocompatibility under accelerated aging conditions.
- Built automated aging experiments using Raspberry Pi, PySerial, and MATLAB, contributing to a conference publication.

## SELECTED RESEARCH PROJECTS

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### Laya: EEG Foundation Model

2024–2026

- Developed Laya, the first JEPA-based EEG foundation model, showing that latent prediction yields more transferable, noise-robust representations than reconstruction-based pretraining.
- Pretrained on 30K hours ( $> 1\text{B}$  tokens) across 20K+ subjects and 17 montages; evaluated on 14 BCI and clinical tasks via frozen linear probing.
- Controlled ablations isolate the pretraining objective as the primary driver of gains over reconstruction baselines.

### Thalamocortical Seizure Dynamics

2023–2026

- Characterized frequency-specific, directed thalamocortical connectivity across the full seizure trajectory in 19 pediatric epilepsy patients using SEEG.
- Showed that slow-frequency thalamic outflow to cortex exceeds cortico-cortical outflow during seizures, with nucleus-specific recruitment matching known anatomy (AN for limbic, CM for frontal).
- Thalamic spectral and connectivity features alone classify seizure state with AUROC 0.83–0.84.

### Seizure World Model

2025–Present

- Building a world model for seizure dynamics that encodes intracranial EEG into a learned latent space and predicts brain-state trajectories conditioned on stimulation parameters.
- Counterfactual rollouts forecast stimulation outcomes for closed-loop neurostimulation planning.

## SKILLS

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|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Neuroscience</b>              | EEG/SEEG preprocessing; spectral and connectivity analysis; neural decoding; brain-computer interfaces; neuroanatomy                 |
| <b>Machine Learning</b>          | State-space models (Mamba); transformers; self-supervised learning; foundation models; time-series modeling; representation learning |
| <b>Programming &amp; Tooling</b> | Python; PyTorch; PyTorch Lightning; MNE; scikit-learn; TensorFlow; SQL; Pandas; Git; distributed training (DDP); Claude Code; Codex  |

Last updated: July 2, 2026